Systematic Characterization of Circumferential Flux Distributions in Parabolic Trough Receivers Using Adaptive Gaussian Representations

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Supplementary Information

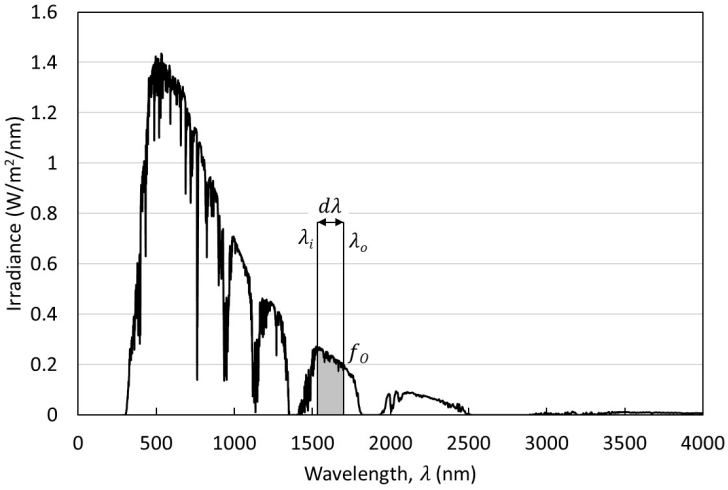


Figure S1: Reference solar spectral irradiance (ASTM G173-03), used for wavelength-resolved MCRT simulation. Reproduced from the ASTM G173-03 direct normal irradiance reference spectrum [1].

Table S1: Statistical indicators for Region 1 candidate models, Jeter's curve at 0° incidence angle.

Model	R^2	RMSE	MAE	SSR	AIC	BIC
Inv.G.	0.9911	0.1638	0.1038	0.16	-13.7	-14.5
DoG.	0.9987	0.0627	0.0467	0.02	-21.2	-22.5

Table S2: Statistical indicators for Region 2 candidate models, Jeter's curve at 0° incidence angle.

Model	R^2	RMSE	MAE	SSR	AIC	BIC
Gaussian	0.9511	3.7347	2.8547	739.258	146.674	151.585
Sup.G.	0.9831	2.195	1.4627	255.35	91.3341	99.215
Skw.G	0.9958	1.0919	0.8376	63.184	17.3152	25.1964
Spt.G	0.9953	1.1516	0.8772	70.282	22.9583	30.8395
Tru.G.	0.9543	3.6086	2.6988	690.165	144.032	151.913
F.G.	0.9511	3.7347	2.8547	739.258	145.674	151.585
Twn.G.	0.9882	1.8342	1.2759	178.307	76.3004	88.122
GMM3	0.999	0.5418	0.409	15.5597	-46.957	-29.224
GMM4	0.999	0.5418	0.409	15.5597	-40.957	-17.314

Full parameter sets underlying all tables in this Supplementary Information are available at `https://github.com/otsun-uib/GaussianLabSuppMat`.

References

[1] ASTM International, Standard tables for reference solar spectral irradiances: Direct normal and hemispherical on 37° tilted surface (astm g173-03) (2003).
URL `https://store.astm.org/g0173-03.html`